

Pablo Francisco Coral — PabloCoral.com

linkedin.com/in/pablo-coral/

EDUCATION

B.S. in Aerospace Engineering, B.M. in Piano Performance
University of Florida, Gainesville, FL

Aug 2022 – May 2027
GPA: 3.47

TOP SKILLS

Design | Manufacturing: Fusion 360, OnShape, Siemens NX, Solidworks, Sketch | ACE CNC Certified, Fusion 360 CAM

Programming | Modeling: TypeScript, Python, MATLAB, C++ (Arduino) | Numerical Methods, Signal Processing, Differential Equations

Front-End | Cloud Infrastructure: React Native, React Expo, Tailwind CSS | AWS, Render, Vercel, Supabase

EXPERIENCE & LEADERSHIP

Manufacturing and Teaching Assistant

Aug 2024 – Present

Design and Manufacturing Lab, UF

Gainesville, FL

- Lead groups of 4 students through a semester-long manufacturing laboratory, teaching them the safe operation of manual machinery including lathes, mills, bandsaws, MIG welders and sheet metal equipment while mentoring teams through the fabrication of 4 compressed-air engines
- Mentor students in Design for Manufacturability and Assembly (DFMA) practices, providing technical guidance and corrections on design reports, air engine part drawings and homework assignments; grade assignments while providing feedback to allow for constant improvement throughout
- Enforce lab safety standards by training students on machine operation and critical safety procedures at the beginning of every semester, monitoring compliance during subsequent lab sessions and maintaining a safe manufacturing environment through consistent application of the safety protocols
- Support lab manufacturing operations by CNC machining components (flywheels, piston blocks) necessary for the students' projects, fabricating replacement parts for damaged or inefficient shop equipment, and applying machine shop experience to keep projects and equipment running
- Stay up to date with the most recent technologies, certificates, machining techniques and safety standards to ensure industry-relevant education for the students, and to be able to assist in day-to-day lab operations with new machinery, tools and general lab organization as dictated by safety organizations

IPT Power Plant & Mech. Systems Engineering Intern

Jan 2025 – May 2025

Piper Aircraft Inc.

Vero Beach, FL

- Designed a custom parking brake valve arm, deriving allowable range of motion from manufacturer datasheets and aircraft firewall constraints, then implemented a formula-driven Siemens NX design compliant with Piper manufacturing standards for rapid design iteration and manufacturability
- Reverse-engineered an aftermarket oleo strut using precise metrology techniques and tools, taking over 200 measurements over 2 weeks to fully model the device in Siemens NX; briefed a team of 5+ engineers on the mechanism of the strut to better understand how to design one of our own
- Developed a Python tool to run parametric simulations of pilot field of view across varying aircraft nose lengths by deriving geometry from a CAD model, performing trigonometric analysis, and using Excel to fit an affine model to the data, establishing a mathematical relationship

INVOLVEMENT

- **Florida Rocket Lab, Manufacturing Engineer** — Fall 2023 – Spring 2025 : Improved manufacturing processes by creating detailed guides and reports for critical components. Used Jira and Confluence (CRM) to create manufacturing-related standard operating procedures (SOPs). Machined 12+ parts for a liquid-propellant rocket engine, trained and supervised new members, and provided guidance about safety practices
- **Florida Rocket Lab, Fluid Systems Engineer** — Fall 2023 – Spring 2025 : Took part in the design of a non-regenerative water-cooling solution for the first version of a liquid-prop. engine. Designed and built a custom scale to accurately measure the mass flow rate of test tanks to ensure operational safety
- **Bike Bros, Partner** — Summer 2025 – Present : Identified and addressed an unmet market for high-quality vintage bicycles at affordable prices. Created a network of 3 bike mechanics to source and repair bikes, and led social media campaigns to match the bicycle with appropriate buyers in need
- **Knead, Lead Operating Officer** — Spring 2026 – Present : Coordinate a group of 5+ engineers towards the shared vision of an investment platform to simplify quantitative investing. Refined and presented the pitch deck and business plan, successfully winning the 2026 Gator Tank Pitch Competition

ENGINEERING PROJECTS

Main Skills — Fusion CAM, Siemens NX, SolidWorks, DFMA | Python, React + TypeScript | Manual Machining

- **Designed and machined** a custom steel pen over 10+ iterations using a manual lathe, independently learning single-point threading techniques
- Programmed, **CNC-machined** and assembled 5 components for an air engine using Fusion and a Haas VF-3 mill, earning the ACE CNC certificate
- Designed a heat-transfer device to **smooth the sinusoidal heating profile of electric stoves**, using SolidWorks for design and Fusion360 for CAM, while abiding by DFM principles and producing a design that is both hand and CNC machinable, using affordable, available parts from McMaster-Carr
- Engineered an automated **Python** pipeline for **Fully Stressed Design of 2D trusses**, integrating custom automation scripts with an already existing **Finite Element Analysis (FEA)** software “ABAQUS”, in order to programmatically run 100+ iteration FSD runs, saving 4+ hours of manual labor

ENGINEERING MODELING & PROGRAMMING PROJECTS

Brake Rotor Temperature Prediction @ Piper Aircraft — Python, Pandas, TensorFlow

- Developed a parametric delay-differential equation (DDE) to predict brake rotor thermal response during ground testing from ~47,000 test sensor readings via mini-batch stochastic gradient descent, and deployed the calibrated model for short-horizon (~2min) thermal forecasting
- Derived and numerically solved the nonlinear DDE using a custom adaptive RK45 integrator, modeling delayed heat input as a function of airspeed and applied brake pressure with temperature-dependent convective dissipation, ensuring numerical stability and real-time updates under dynamic loading

Regime-Adaptive Bollinger Bands Algorithm — Python, Pandas, NumPy, hmm-learn, Alpaca-Py

- Designed and deployed a **5-state Hidden Markov Model** to classify latent market volatility regimes using log returns and intra-bar range features
- Tested the regime-adaptive strategy against static-band & buy-and-hold baselines under transaction costs with walk-forward validation; improved risk-adjusted performance relative to fixed bands but underperformed passive trend in sustained bull regimes, showing the strategy's regime-specific edge